

Mg-CO₂ Electrochemical System: A Study on Bubble Diameter and Reaction Kinetics Effects

Raman Kumar Singh, Prakash C Ghosh, Venkatasailanathan Ramadesigan

Department of Energy Science and Engineering, Indian Institute of Technology Bombay

Mumbai, Maharashtra, India, 400076

ramanras@iitb.ac.in

The continuous increase in atmospheric carbon dioxide (CO₂) concentrations since the Industrial Revolution have led to various environmental impacts, including rising sea levels, unpredictable flooding and droughts, and global warming. Recent advancements in electrochemical CO₂ conversion presents a promising solution for reducing CO₂ emissions while enabling the sustainable production of hydrogen and value-added fuels. CO₂ gas dissolves in water, forming HCO₃⁻, H⁺, and CO₃²⁻ ions, which lowers the electrolyte pH. On this concept, a 3-D multiphase transient Mg-CO₂ electrochemical model is developed to utilize CO₂ into hydrogen Mg(HCO₃)₂ and electricity. The dilute solution theory to explain species transport, the Euler-Euler (EE) multiphase approach for the conversion of gaseous CO₂ to aqueous CO₂, and the porous electrode theory for the electrochemical conversion of H₂ within porous media is used to develop the model, and validated with the experimental data from Kim et al. [2]. The bubble diameter significantly influences the gas-liquid interfacial area, impacting the efficiency of mass transfer in the Mg-CO₂ system. Changes in the chemical rate constant further affect the reaction kinetics, playing a critical role in system performance. A detailed analysis of these parameters is done to improve the CO₂ conversion efficiency of the Mg-CO₂ system.

References

- [1] Ishaq, Haris, et al. "A review on hydrogen production and utilization: Challenges and opportunities." *International Journal of Hydrogen Energy* 47.62 (2022): 26238-26264.
- [2] Kim, Jeongwon, et al. "Indirect surpassing CO₂ utilization in membrane-free CO₂ battery." *Nano Energy* 82 (2021): 105741.